Sergio Fuentes Cámara

Computer Vision Engineer - ML Edge Computing Specialist

Madrid, España   +34 666 113 180   s.fuentescamara@gmail.com
Portfolio: myprofolio-mwfs.onrender.com   Github   LinkedIn

Professional Profile

Computer Vision Engineer with +5 years specialising in real-time AI systems for edge devices. Experience implementing detection, OCR and tracking models in production processing +20 FPS in Full HD. Stack: Python, TensorFlow, PyTorch, Docker, FastAPI.

Professional Experience

Computer Vision Engineer — Grupo Etra *2023 - Present*

- Architected and implemented end-to-end Computer Vision pipeline in C++/Python processing Full HD video in real-time (<50ms latency) on NPU edge devices, reducing hardware costs by 70% vs. GPU solutions
- Designed and optimised custom object detection and OCR models achieving 97% accuracy in production, processing +15K operations/day through fine-tuning and transfer learning
- Led DevOps infrastructure for fleet of +20 edge devices implementing CI/CD with Docker, Ansible, and centralized monitoring, reducing deployment time
- Developed full-stack web platform (FastAPI, WebSocket, WebRTC) managing +10 simultaneous streams in real time, reducing remote configuration time by 5x

Software Engineer — Capgemini España SL *2022 - 2023*

- Developed Computer Vision system for industrial quality control automating metric measurement in production line, improving precision and reducing inspection times
- Implemented CI/CD pipelines with DevOps methodology (Git, Docker, Jenkins) for aeronautical project, ensuring continuous delivery and reducing production errors
- Designed scalable and modular software architectures following SOLID principles and design patterns, facilitating code maintainability and extensibility
- Built data preprocessing pipelines with NumPy and Pandas to feed ML models, optimizing performance and prediction quality

Electronics Engineer and Firmware Developer — Digitanimal *2020 - 2022*

- Designed and programmed low-power microcontroller firmware in C for agricultural IoT products, optimizing battery autonomy and GSM/LoRa/Sigfox communication protocols
- Developed sensor processing algorithms (GPS, accelerometer, temperature) and efficient data transmission for real-time livestock monitoring systems
- Designed complete electronic circuits including schematics, PCBs, and prototype validation for GPS collars and self-weighing scales in harsh industrial environments
- Led full prototyping cycle from mechanical 3D design (SolidWorks) to manufacturing with 3D printing (FDM/resin), rapidly iterating to final product
- Contributed to R&D of new IoT products evaluating technical feasibility, developing POCs, and accelerating time-to-market for hardware-software innovations

Technical Skills

Computer Vision & AI:

- TensorFlow & PyTorch • Object detection & Tracking • OpenCV • OCR • Model Optimization

Edge Computing & Hardware:

- NPU acceleration • Low-latency systems • Embedded C/C++ • Microcontrollers & IoT

Backend & APIs:

- Python
- FastAPI & REST APIs
- WebSockets & WebRTC
- MongoDB & SQL
- Asyncio & multiprocessing

DevOps & Cloud:

- Docker
- Ansible
- CI/CD (Git, Jenkins)
- AWS & GCP
- Linux & Bash

Frontend & Design:

- JavaScript & HTML/CSS
- React
- Flutter
- SolidWorks (3D design)
- 3D Printing (FDM/Resin)

Languages

Spanish : Native

English : Technical (documentation, papers, code) — Conversational in development

Education

Bachelor's Degree in Industrial Electronics and Automation Engineering

University of Castilla-La Mancha (UCLM), Toledo

Specialization in Advanced Electronic Technologies

Certificates and Courses

TensorFlow Developer, *DeepLearning.AI* (link) — *August 2023*

DevOps Fundamentals Course, *GitHub/90DaysOfDevOps* (link) — *October 2022*

AWS Computer Vision with GluonCV, *AWS Training and Certification* (link) — *July 2022*

Image Segmentation with Python and Unsupervised Learning, *Coursera Project Network* (link) — *July 2022*

Object Detection in Computer Vision, *Autonomous University of Barcelona* (link) — *January 2020*